

Insights

Inventory Management & Cross-Border Supply Chain Analysis

Summary

This document presents the key findings from the Inventory Management & Cross-Border Supply Chain Analysis Power BI dashboard, connecting each observed result to its underlying driver and a corresponding business recommendation. Findings are drawn from analysis of 25,000 records across 5 regions, 15 countries, 5 product categories, and 4 warehouses spanning January 2023 to December 2024. All KPI calculations and assumptions underpinning these findings are documented in the accompanying Methodology & Assumptions report.

Finding 1: Profit Margin is Strong and Consistent Across Both Years

The business demonstrates strong and stable profitability across the entire two-year analysis period. Overall, Net Profit Margin stands at 39.56%, with 2023 recording 39.57% and 2024 recording 39.56%. This consistency is as significant as the margin level itself: it indicates that cost management, pricing discipline, and operational efficiency have been maintained without meaningful deterioration despite the cross-border complexity of operating across 5 regions, 15 countries, and 4 warehouses simultaneously. A margin of 39.56% is healthy by cross-border e-commerce standards, where logistics costs, tariffs typically compress margins significantly.

Finding 2: Beauty Generates the Most Profit Despite the Lowest Inventory Turnover — Challenging the Assumption That Fast Turnover Drives Profitability

One of the most counter-intuitive findings in this analysis is the relationship between inventory turnover rate and profitability across product categories. Beauty has the lowest inventory turnover rate at 1.91 - meaning it converts stock to sales roughly half as fast as Electronics, which leads the network at 3.50. Yet Beauty generates the highest net profit of any category at \$0.927bn with a net profit margin of 58.70%, while Electronics generates the lowest net profit at \$0.391bn with a margin of just 24.86%. This means Beauty generates more than double the profit of Electronics despite turning its inventory at nearly half the speed. The explanation lies in the gross margin structure: Beauty operates at a 65% gross margin, meaning each unit sold retains significantly more value even after cost of goods, while Electronics operates at a 35% gross margin, leaving less room for profit per unit regardless of how quickly inventory moves. This finding challenges a common assumption in supply chain management that faster inventory turnover is always better - in reality, a high-margin, slower-turning category can significantly outperform a fast-turning, thin-margin one.

Finding 3: Delivery Performance is Critically Low and Lead Time is the Root Cause

On-Time Delivery Rate across the network is 22.89%, meaning more than three in every four shipments arrive late. This represents a significant customer satisfaction risk - late deliveries directly erode trust, increase return rates, and reduce repeat purchase likelihood in cross-border e-commerce where customers already accept longer wait times as a baseline. Analysis of the underlying data reveals that this is not a random execution failure or carrier-handling issue - it is structurally driven by lead time. Shipments marked as on-time average just 8.54 days of lead time, while shipments marked as late average 21.42 days. The network-wide average lead time sits at 18.47 days - already close to the threshold that separates on-time from late deliveries - meaning the majority of shipments are structurally positioned to miss the on-time window before they even leave the warehouse. Critically, this is not a regional problem: average lead time across all five regions ranges only from 18.35 to 18.58 days, confirming the issue is systemic across the entire network rather than concentrated in a specific geography. This means fixing one region or one route will not resolve the problem - lead time reduction needs to be addressed as a network-wide operational priority.

Finding 4: Air Shipping Costs Three Times More Than Sea with No Meaningful Speed or Reliability Advantage

A significant cost efficiency gap exists in the current shipping mode mix. Average shipping cost per shipment for Air is \$451.70, compared to \$249.31 for Ground and \$152.88 for Sea - meaning Air costs nearly three times more per shipment than Sea. Despite this substantial cost premium, Air shipping delivers no meaningful advantage in either lead time or on-time delivery performance: Air averages 18.34 days lead time with a 24% on-time rate, Ground averages 18.62 days with a 22% on-time rate, and Sea averages 18.45 days with a 23% on-time rate. All three modes perform virtually identically across both delivery speed and reliability metrics. In addition to cost, Air shipping generates nearly three times the carbon emissions of Sea (360.28 vs 122.50 average units), adding an environmental cost to the financial one. The business case for Air shipping typically rests on its speed advantage, but since that advantage does not materialize in this network's data, the current volume of Air shipments represents unnecessary cost and carbon emission without a compensating performance benefit.

Finding 5: WH_4 Has the Highest Stockout Rate in the Network While All Warehouses Serve the Same Markets

Stockout Rate varies meaningfully across the four warehouses: WH_4 records the highest stockout rate at 2.87%, followed by WH_1 at 2.70%, WH_2 at 2.27%, and WH_3 at the lowest at 2.19%. The gap between WH_4 and WH_3 is 31% - a significant operational difference. What makes this finding particularly important is that all four warehouses serve all five regions with nearly equal transaction volumes, meaning demand-side differences cannot explain the gap. WH_4 and WH_1 are not serving higher-demand or more volatile markets than WH_3; they are operating under similar demand conditions but producing worse stockout outcomes. This points to a warehouse-specific operational issue, likely in replenishment timing, stock allocation, or inventory planning practices at WH_4 specifically - rather than an external demand or supply factor.

Finding 6: Marketing Spend Tracks Market Size Rather Than Profitability — Efficiency is Flat Across All Markets

Countries with the highest marketing spend- Brazil (\$13.4M), Chile (\$13.1M), and Saudi Arabia (\$13.0M) - also record the highest net profits (\$319M, \$308M, and \$318M respectively), which at first glance suggests that higher marketing investment drives higher returns. However,

deeper analysis reveals that this relationship is driven by market size rather than marketing effectiveness. These countries have proportionally more transactions in the dataset, and when marketing spend is expressed as a percentage of revenue, it ranges only from 1.62% to 1.73% across all 15 countries - an extremely narrow band that indicates spend is being allocated proportionally to revenue scale rather than concentrated in any particular market. Net Profit Margin is equally flat, ranging from 38.99% to 40.61% across all markets, with no country showing meaningfully higher or lower profitability regardless of spend level.